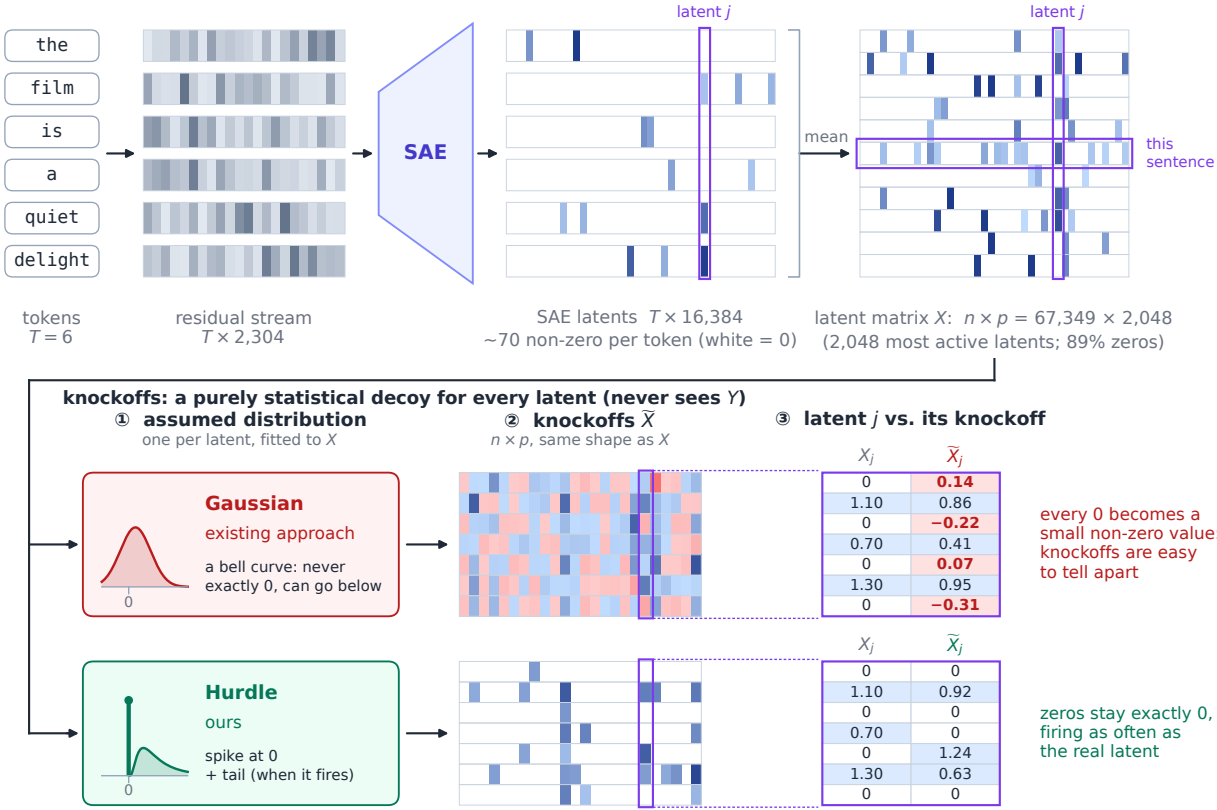

the
film
is
a
quiet
delight
SAE
latent j
mean
latent j
this sentence
tokens
T = 6
residual stream
T × 2,304
SAE latents T × 16,384
~70 non-zero per token (white = 0)
latent matrix X: n × p = 67,349 × 2,048
(2,048 most active latents; 89% zeros)
knockoffs: a purely statistical decoy for every latent (never sees Y)
① assumed distribution
one per latent, fitted to X
② knockoffs X̃
n × p, same shape as X
③ latent j vs. its knockoff
Gaussian
existing approach
a bell curve: never exactly 0, can go below
0
Hurdle
ours
spike at 0
+ tail (when it fires)
0
X_j
X̃_j
0 0.14
1.10 0.86
0 −0.22
0.70 0.41
0 0.07
1.30 0.95
0 −0.31
every 0 becomes a small non-zero value: knockoffs are easy to tell apart
X_j
X̃_j
0 0
1.10 0.92
0 0
0.70 0
0 1.24
1.30 0.63
0 0
zeros stay exactly 0, firing as often as the real latent